

Approximation algorithms

Problem Π , minimize the cost or maximize the benefit
(NP-hard)

optimal solution S^* has $\text{cost}(S^*) = \text{OPT}$

Def. An r -approximation, $r > 1$, gives a solution S such that

minimization

$$\frac{\text{cost}(S)}{\text{OPT}} \leq r$$

maximization

$$\frac{\text{OPT}}{\text{cost}(S)} \leq r$$

[sometimes, it is equivalently
called $\frac{1}{r}$ -approximation
as $\text{cost}(S) \geq \frac{1}{r} \text{OPT}$]

We are interested in poly-time approximations
(whenever possible)

We will use KNAPSACK as an illustrative example

IN: n items, item i $\begin{cases} \rightarrow \text{value } v_i > 0 \\ \rightarrow \text{weight } w_i > 0 \end{cases}$ $(1 \leq i \leq n)$, capacity $W > 0$

OUT: choose $S \subseteq [n]$ s.t. $\begin{cases} \rightarrow \sum_{i \in S} w_i \leq W & (\text{items in } S \text{ should fit}) \\ \rightarrow \text{maximizes } \sum_{i \in S} v_i & (\text{total value of items in } S) \end{cases}$

- we first see a 2-approximation

- we discuss how to get a FPTAS (i.e. any r -approx $r > 1$)

2-approximation (greedy)

Intuition: give priority to most valuable items, that is, $\frac{V_i}{W_i}$ ratio

Hp. wlog. suppose items are numbered so as

$$\textcircled{1} \quad \frac{V_1}{W_1} \geq \frac{V_2}{W_2} \geq \dots \geq \frac{V_n}{W_n} \quad (\text{o.w. sort and renumber items})$$

$$\textcircled{2} \quad S = \{\}$$

for $i = 1, 2, 3, \dots, n$: (ordered as $\textcircled{1}$)

if $w_i \leq W$ then $S = S \cup \{i\}$

$$W = W - w_i$$

return S'

We denote by $m_{\text{greedy}} = \text{cost}(S) = \sum_{i \in S} v_i$

Note that this GREEDY does not guarantee the OPT value

$$\left. \begin{array}{l} v_1 = v_2 = \dots = v_{n-1} = 1 \\ w_1 = w_2 = \dots = w_{n-1} = 1 \end{array} \right\} \frac{v_i}{w_i} = 1$$

$$\left. \begin{array}{l} v_n = W-1 \\ w_n = W \end{array} \right\} \frac{v_n}{w_n} < 1, \text{ where } W = kn \text{ for any large } k > 1$$

$$\text{GREEDY } m_G = n-1 = \sum_{i \in S} v_i, \text{ as } S = \{1, 2, \dots, n-1\}$$

$$\text{OPT} = W-1$$

$$\text{Thus, } \frac{\text{OPT}}{\text{cost}(S)} = \frac{W-1}{n-1} = \frac{kn-1}{n-1} \sim k \text{ for any large } k > 1$$

it does not either guarantee a 2-approx

To fix this issue, we add a third step:

(3) return $\max \{m_G, V_{\max}\}$, where $V_{\max} = \max_{1 \leq i \leq n} V_i$

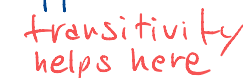
We now show that this variation now guarantees a 2-approximation

Implicit argument in any approximation alg. (known to the teacher) 

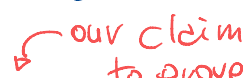
Since we do not know OPT, how can we prove things?

It suffices to have an upper bound $UB \geq OPT$: indeed

maximization

transitivity as $OPT \leq UB$ 

$$\frac{OPT}{cost(S)} \leq \frac{UB}{cost(S)} \leq r$$

our claim to prove 

transitivity as $LB \leq OPT$

minimization

$$\frac{cost(S)}{OPT} \leq \frac{cost(S)}{LB} \leq r$$



We need to find an upper bound to OPT

- ① if all items fit into W , we have OPT. easy case
- ② otherwise, some items should be discarded

GOAL:

$$\frac{\text{OPT}}{\underbrace{\max\{m_G, V_{\max}\}}_{\text{Cost}(S)}} \leq 2$$

First, let j be the first item that does not fit

$$\sum_{i=1}^{j-1} w_i \leq W \quad \text{but} \quad \sum_{i=1}^j w_i > W$$

$$\underbrace{\hspace{1.5cm}}_{\bar{w}_j}$$

$$\text{and } \bar{V}_j = \sum_{i=1}^{j-1} V_i$$

largest prefix of
consecutive items
that fit
(OPT does the same)

CLAIM :

$$\text{OPT} < \bar{V}_j + V_j$$

our upper bound UB

▶ Suppose that item j can be split into j' and j''
such that $V_j = V_{j'} + V_{j''}$, $W_j = W_{j'} + W_{j''}$ and $W_{j'}$ fits perfectly
what left
then we get OPT ∇

$$\left(\sum_{i=1}^{j-1} w_i + w_{j'} = W \right)$$

Two cases :

steps ①+②

$$\blacklozenge \bar{V}_j > V_j \Rightarrow \text{OPT} < 2 \bar{V}_j \leq 2 m_G \leq 2 \max \{ m_G, V_{\max} \}$$

steps ③

$$\blacklozenge \bar{V}_j \leq V_j \Rightarrow \text{OPT} < 2 V_j \leq 2 V_{\max} \leq 2 \max \{ m_G, V_{\max} \}$$

We can equivalently write:

$$\text{OPT} \cdot \frac{1}{2} \leq \text{cost}(S)$$

Natural question: $\text{OPT} \cdot (1-\epsilon) \leq \text{cost}(S)$?

FPTAS (i.e. poly $\frac{W}{1/\epsilon}$)

$$0 < \epsilon < 1$$

HANDS-ON: suppose that $w_i \leq \epsilon W$

Prove that steps ① + ② give

Hint: the first time j does not fit, the knapsack is at least $(1-\epsilon)$ full

Now... FPTAS in the general case: any $w_i \in W$ and any $0 < \epsilon < 1$

$$\boxed{\text{OPT} \cdot (1 - \epsilon) \leq \text{cost}(S)}$$

the smaller ϵ , the better is

running time is $O(n^3/\epsilon)$

Recall our DP2 solution in $O(n^2 V_{\max})$ time: EXACT

IDEA: Since $V_{\max}$ can be very large, we scale all the values

$$v_i \rightsquigarrow \tilde{v}_i = \left\lfloor \frac{v_i}{K} \right\rfloor \text{ with } K \text{ to be fixed}$$

(A) ORIGINAL INSTANCE: v_i, w_i, W

(B) SCALED INSTANCE: $\tilde{v}_i, w_i, W$ + DP2 exact solution

Fact

◆ S is a solution for (B) $\Rightarrow S$ is a solution for (A)
(weights do not change)

◆ $\tilde{V}_{\max} = \left\lfloor \frac{V_{\max}}{k} \right\rfloor$

Clearly, the cost changes!

GOAL: choosing $k = \frac{\epsilon V_{\max}}{n}$, the optimal solution S_{in} (B) satisfies

$$\sum_{i \in S} V_i \geq (1-\epsilon) \text{OPT} = (1-\epsilon) \sum_{i \in S^*} V_i$$

note: it's not $\tilde{V}_i \dots$

we are using S as a solution in (A)!

Fact Recall $\tilde{v}_i = \left\lfloor \frac{v_i}{k} \right\rfloor$. Hence..

$$\frac{v_i}{k} - 1 \leq \tilde{v}_i \leq \frac{v_i}{k} \quad \text{because of } \lfloor \dots \rfloor. \text{ Thus, multiplying by } k$$

$$(i) \quad v_i \geq k \tilde{v}_i$$

$$(ii) \quad k \tilde{v}_i \geq v_i - k$$

Let's prove the GOAL

Since S is optimal in (B) and S^* is optimal in (A)

S^* cannot be better than S in (B)

$$(iii) \quad \underbrace{\sum_{i \in S} \tilde{v}_i}_{\text{optimum in (B)}} \geq \sum_{i \in S^*} \tilde{v}_i$$

$$\sum_{i \in S} v_i \geq \sum_{i \in S} k \tilde{v}_i \geq \sum_{i \in S^*} k \tilde{v}_i \geq \sum_{i \in S^*} (v_i - k) = \underbrace{\sum_{i \in S^*} v_i}_{\text{OPT}} - |S^*| \cdot k$$

$$\geq \text{OPT} - kn \geq \text{OPT} - \epsilon \text{OPT} = (1 - \epsilon) \text{OPT}$$

$$|S^*| \leq n$$

$$V_{\max} \leq \text{OPT}$$

To complete this proof; $kn = \epsilon V_{\max} \leq \epsilon \text{OPT}$

QED

our choice $k = \frac{\epsilon V_{\max}}{n}$

Running time: $O(n^2 \tilde{V}_{\max}) = O(n^3 / \epsilon)$

$$\hookrightarrow \tilde{V}_{\max} = \left\lceil \frac{V_{\max}}{k} \right\rceil \leq \frac{V_{\max}}{k} = \cancel{V_{\max}} \cdot \frac{n}{\epsilon \cancel{V_{\max}}} = \frac{n}{\epsilon}$$